

PUF+SNN Security for Quest 3 Motion Data

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Baseline Setup

- 1,800 synthetic motion windows
- 600 train, 600 validation, 600 test
- Sessions 1, 2, and 3 define the splits
- 120 time steps and 840 features
- Logistic regression and random forest

Primary split cross-session

Session 1 train → Session 2 validation → Session 3 test

The same synthetic device profiles appear in every split

5 classes: nod, shake, look left, look right, still

Random seed: 2026

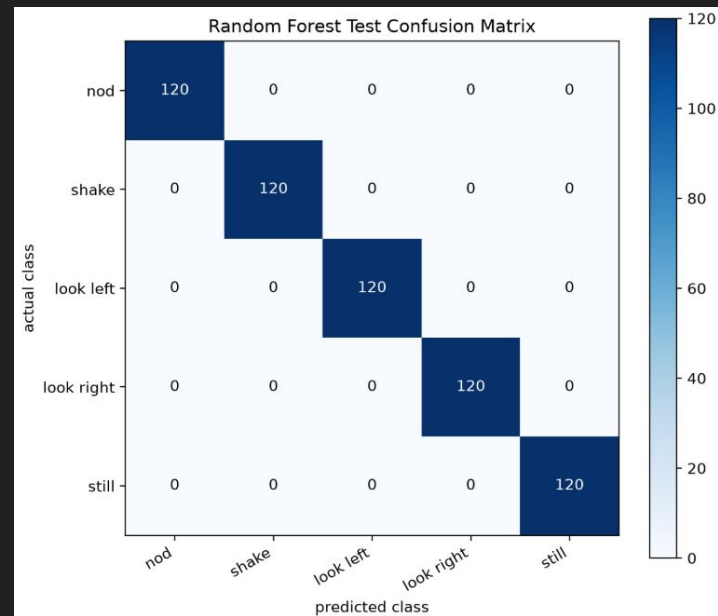
Baseline Results

All five classes: precision, recall, and F1 = 1.0000

Provisional post-window target: 20 ms p95

Model	Validation macro-F1	Test accuracy	Test Macro-F1	Median	p95
Logistic Regression	1.0000	1.0000	1.0000	0.2322 ms	0.39 ms
Random Forest	1.0000	1.0000	1.0000	14.1423 ms	21.65 ms

Both models produced the same confusion matrix



Quest Logger



Window Checks

- Position, orientation, timestamp, and tracking
- Timestamp gap < 50 ms
- Valid tracking > 95%

Trial timing: 1 s prompt, 2 s capture, 1 s rest

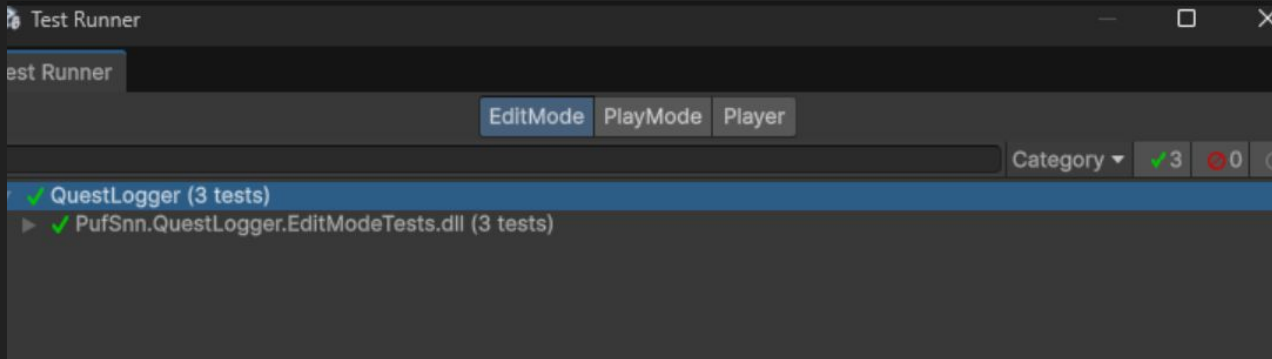
Validation & Next Steps

Validation

- 3 of 3 EditMode tests passed
- 0 OpenXR validation issues
- 0 compilation errors
- Human recording remained disabled

Next steps

- Connect the Quest through ADB
- Prepare the Tier 2 abnormality design before training the detector
- Build and install the application
- Run an approved test capture
- Pull and validate the JSONL file



Repository: github.com/Keellonn/PUF-SNN, Baseline run commit: 4f540a7, Logger implementation commit: d2f613c

PUF Baseline: Configuration and Numerical Results

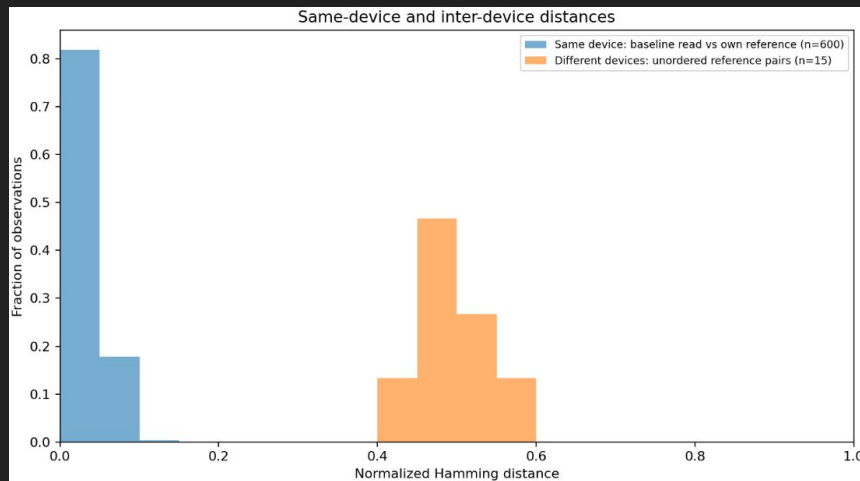
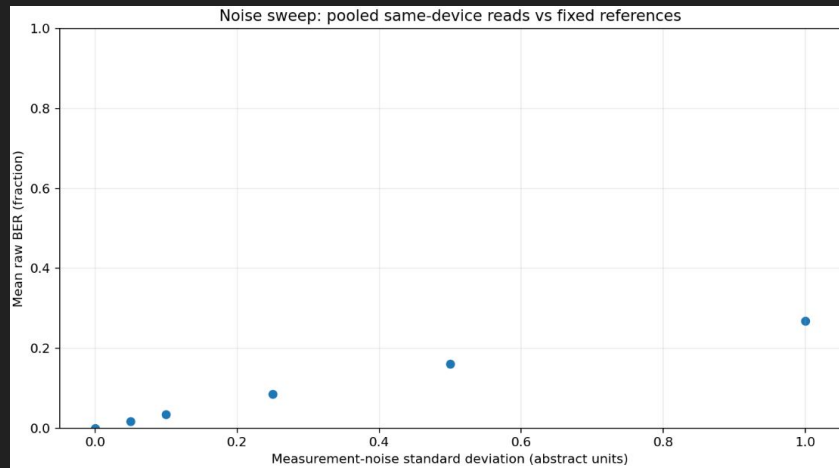
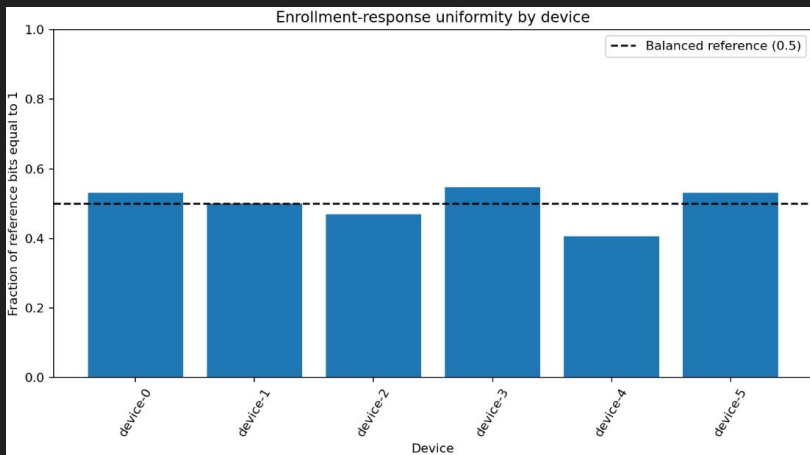
- 20 independently seeded simulations
- 6 devices/run, 128 ROs/device
- Adjacent pairing → 64-bit response
- 100 repeated reads/device
- Nominal measurement noise = 0.1
- Mean BER, 3.2076%
- Reliability: 96.7924%
- Uniqueness: 49.5781%
- Uniformity: 50.0651%

Metric	Mean	Standard Deviation	Min	Max
Raw BER	0.032075521	0.004237424	0.022630208	0.042317708
Reliability	0.967924479	0.004237424	0.957682292	0.977369792
Uniqueness	0.49578125	0.016093045	0.45625	0.526041667
Uniformity	0.500651042	0.02785459	0.450520833	0.557291667

Manufacturing Variation	Raw BER	Reliability	Uniqueness
0.25	0.113072917	0.886927083	0.477083333
0.5	0.056640625	0.943359375	0.477083333
0.75	0.037447917	0.962552083	0.477083333
1.00 (Baseline)	0.027734375	0.972265625	0.477083333
1.5	0.019296875	0.980703125	0.477083333
2	0.014505208	0.985494792	0.477083333

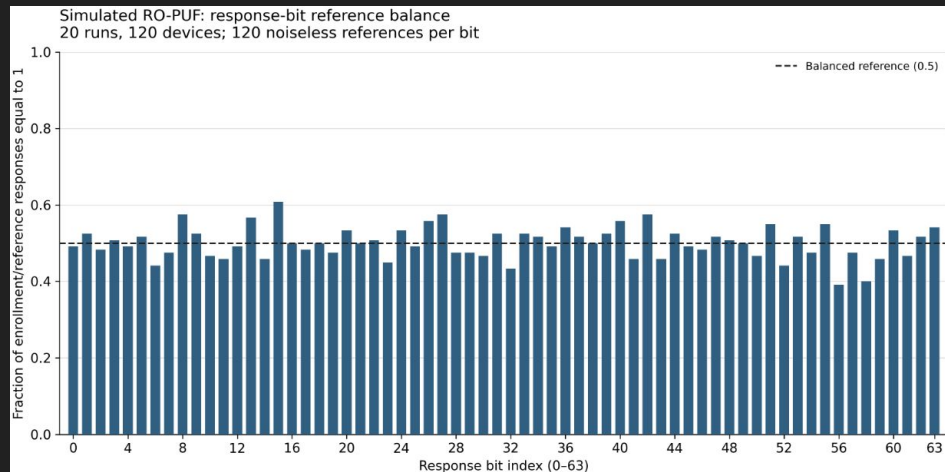
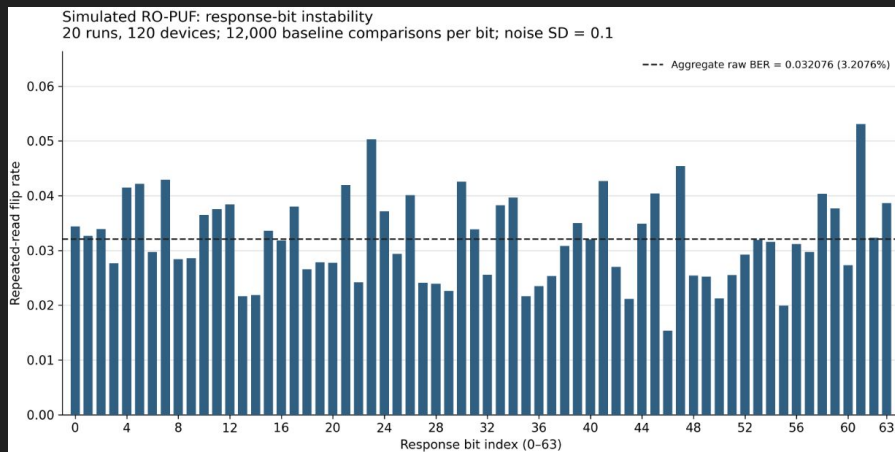
PUF Experimental Results

- Same-device responses cluster near enrollment reference
- Different-device responses differ in roughly half their bits
- BER increases with measurement noise
- Uniformity remains near 50%

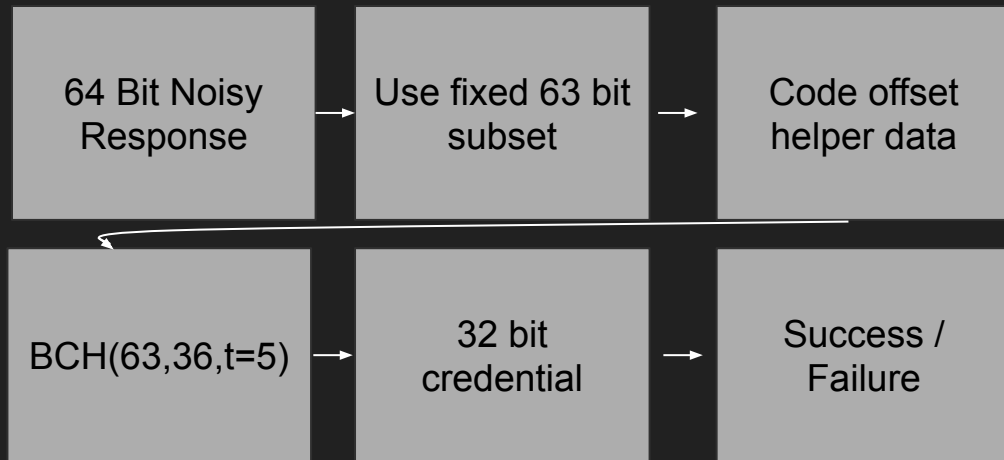


PUF Experimental Results (Continued)

- Per-bit flip rates: **1.53%–5.31%**
- Aggregate per-bit flip rate matches BER: **3.207552%**
- Increasing manufacturing variation improves reliability under fixed noise



Credential-Reconstruction Design



- 32-bit pilot credential
- 63 of 64 PUF bits used for reconstruction
- Candidate BCH(63,36,t=5)
- Code-offset/helper-data construction
- Corrects up to 5 bit errors
- ~2.02 errors expected on average at measured BER
- Actual reconstruction success + FRR still need to be measured

Next Steps

- Implement proposed reconstruction design
- Measure:
 - reconstruction success
 - false-rejection rate
 - reconstruction latency
 - success vs raw BER
- Validate/refine BCH parameters from measured results
- Derive per-session key only after reconstruction succeeds
- HMAC-SHA-256 protect sensor windows
- Begin replay/modification/cross-device Tier-1 testing